

Wear-It-Virtually -- Experiment Report v2

Improved Pipeline | 2026-05-01 | Python 3 | GrabCut Segmentation + Poisson Blending

Key Improvements over v1

Clothing Segmentation	GrabCut extracts clothing from white background, removing fake edges
Perspective Warp	Clothing is perspective-transformed to match body proportions
Poisson Blending	cv2.seamlessClone creates natural color transitions at clothing edges
Background Removal	Alpha mask refined by detecting background-colored pixels

Comparison: Original vs Improved



Original v1: whole clothing image pasted with Gaussian-blur edges (unrealistic borders). Improved v2: GrabCut removes background, Poisson blending creates natural color fusion.

Intermediate Steps (person1 + clothing1)



Person input



Clothing input



Segmentation + Alpha mask



Final result

All Results -- 3 Person-Clothing Pairs



Row 1: person1.jpg (man, front-facing) | Row 2: person2.jpg (woman, slight angle) | Row 3: person3.jpg (woman, different pose) | Each column: Person -- Clothing -- Virtual Try-On Result

Technical Details

Step	Algorithm	Parameters	Output
1. Clothing Segmentation	GrabCut (cv2.grabCut)	5 iterations, rect=(10%,10%,80%,80%)	clothing_segmented.png (RGBA)
2. Background Refinement	Color distance from corner avg	threshold=0.15 (normalized)	refined alpha mask

3. Body Detection	GrabCut on person image	5 iterations, center rect	body_mask.png + bounding box
4. Perspective Warp	cv2.getPerspectiveTransform	4-point src/dst mapping	warped to body region size
5. Edge Morphology	cv2.dilate + cv2.erode	5x5 + 3x3 kernel, 1 iteration each	smooth alpha edges
6. Fusion	cv2.seamlessClone (NORMAL_BLEND)	blending at body center	final.png

Wear-It-Virtually Improved Pipeline -- Python 3, OpenCV (cv2.grabCut + cv2.seamlessClone). Demo mode runs without model weights. Full pipeline: see README.